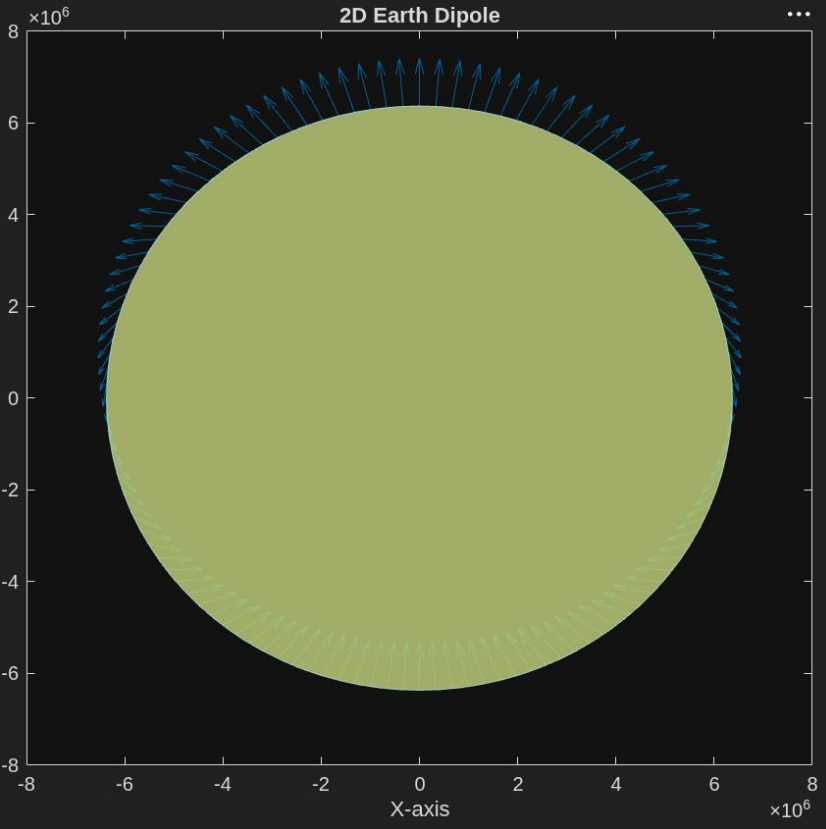
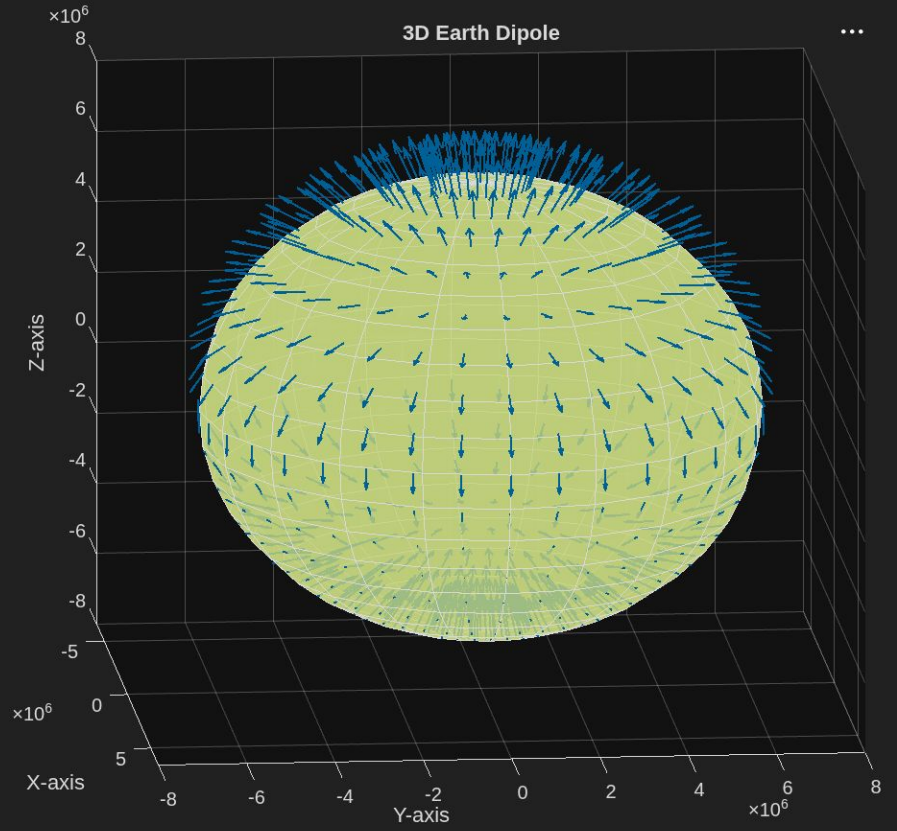
Magnetic Detumbling Simulation

Project by Alexander supervised by Zac Isaac

Magnetosphere Simulation - Dipole

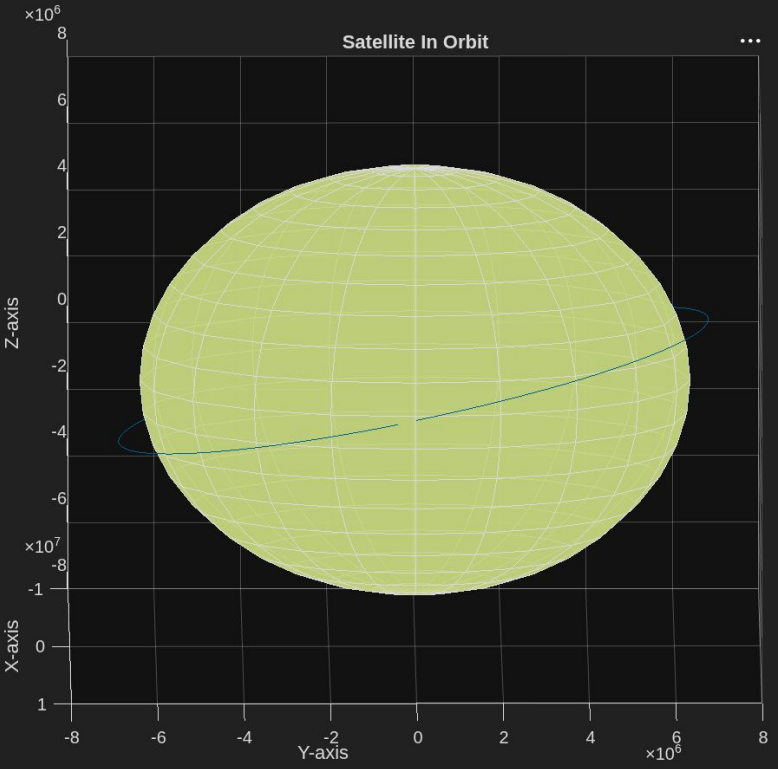





```
polar_step = deg2rad(3);
azimuth_step = deg2rad(3);
```

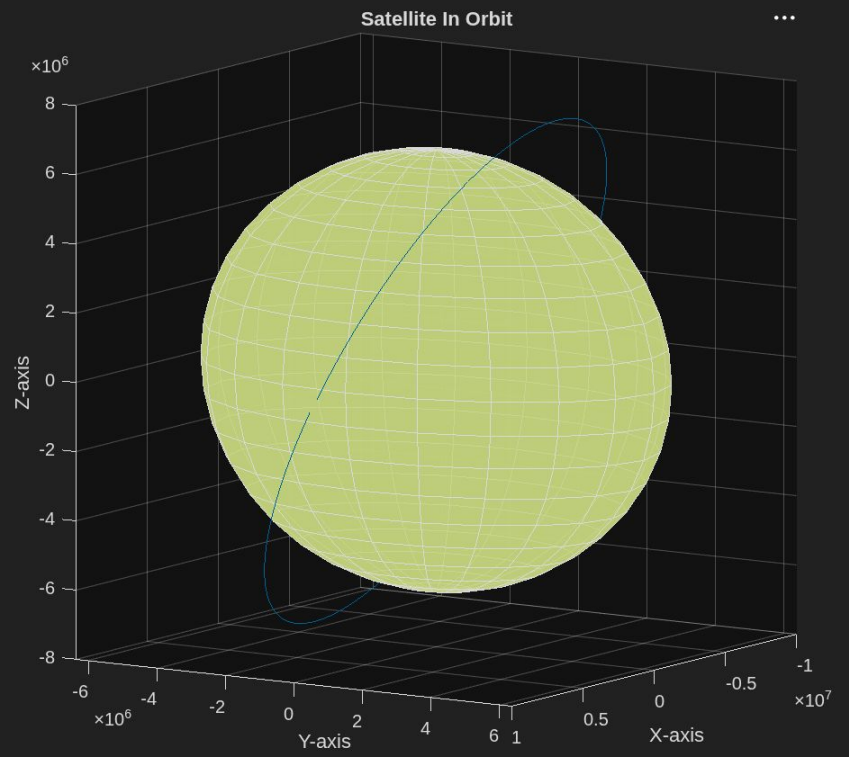
```
polar_step = deg2rad(10);
azimuth_step = deg2rad(10);
```

Satellite Orbit




```
semi_major = 7021e3; % Meters
inclination = deg2rad(15); % Radians
```

Altitude: 650km

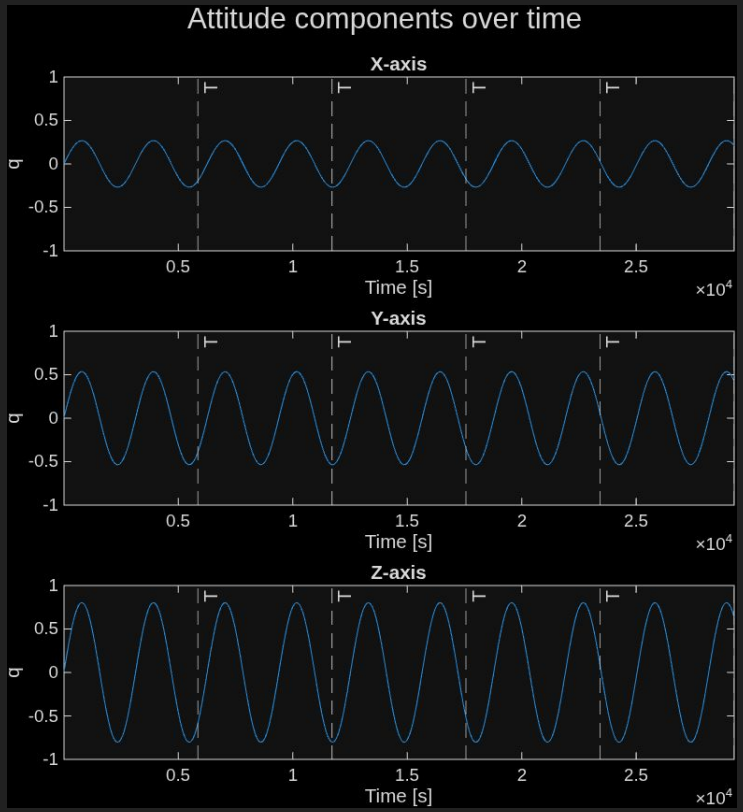


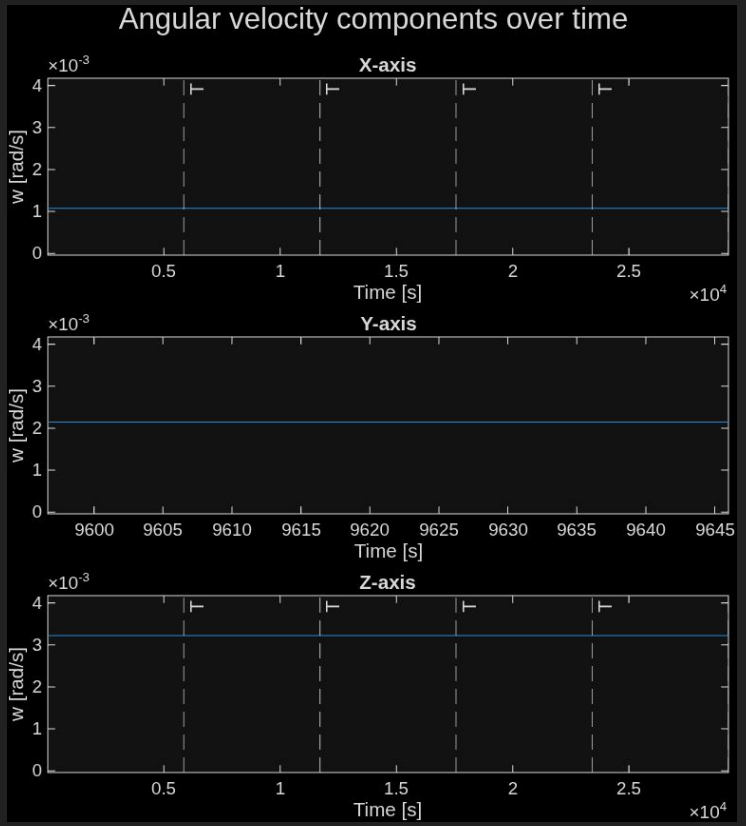

```
semi_major = 8371e3; % Meters
inclination = deg2rad(65); % Radians
```

Altitude: 2000km

Tumble

```
sat.Angular_Velocity = [sat.Mean_Motion sat.Mean_Motion*2 sat.Mean_Motion* 3];
```

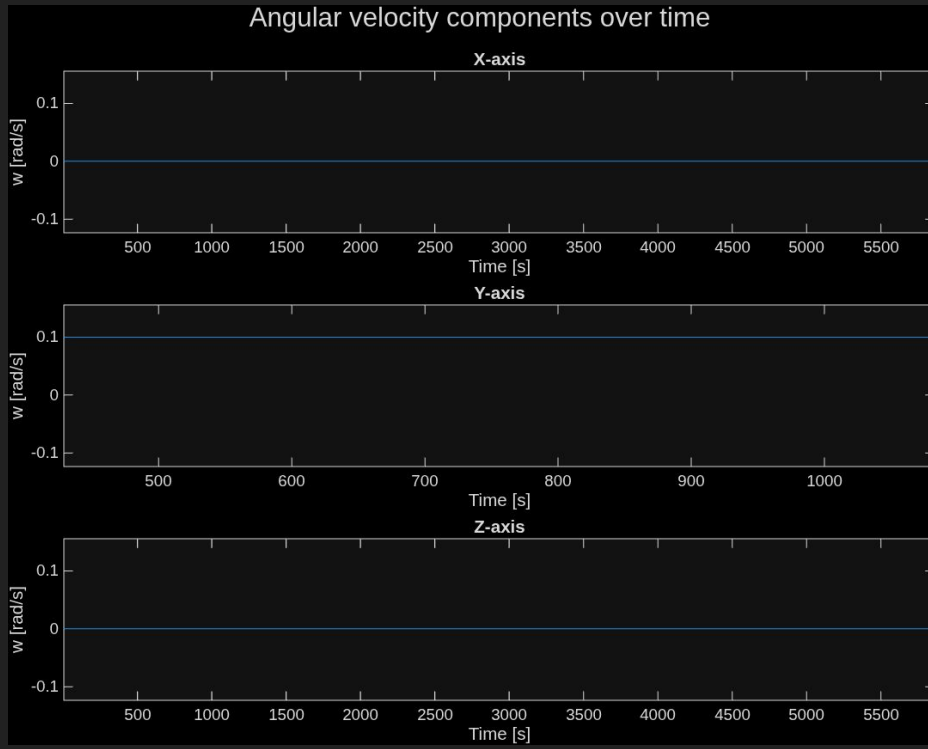


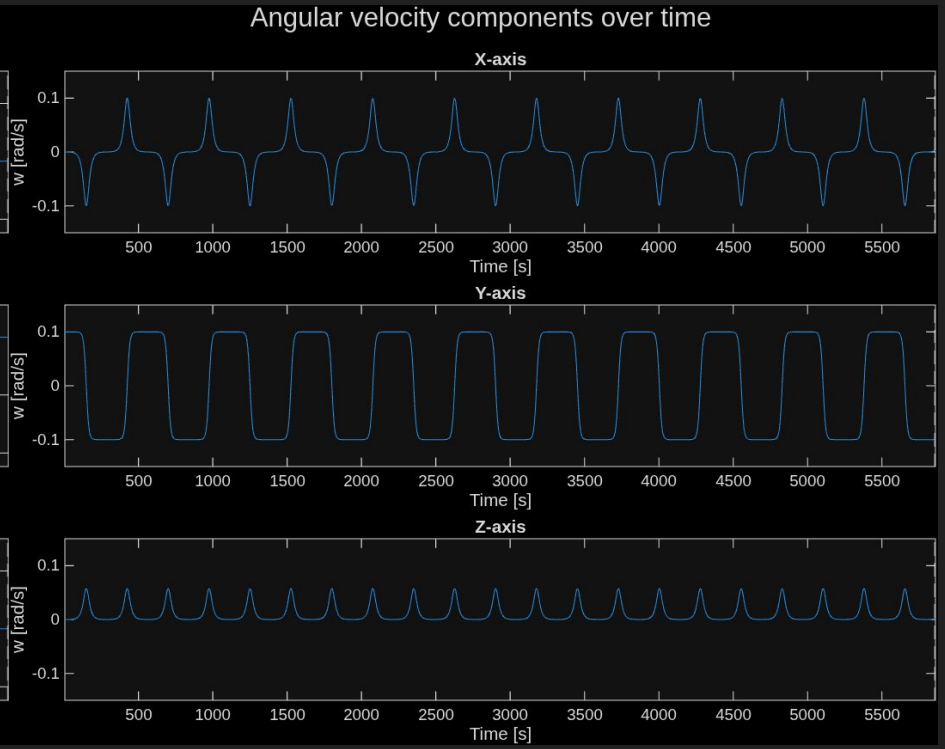


Tennis-racket Effect

```
angular_velocity = [1e-4, 0.1, 1e-4];
inertia_tensor   = diag([0.4 0.4 0.4]);
```

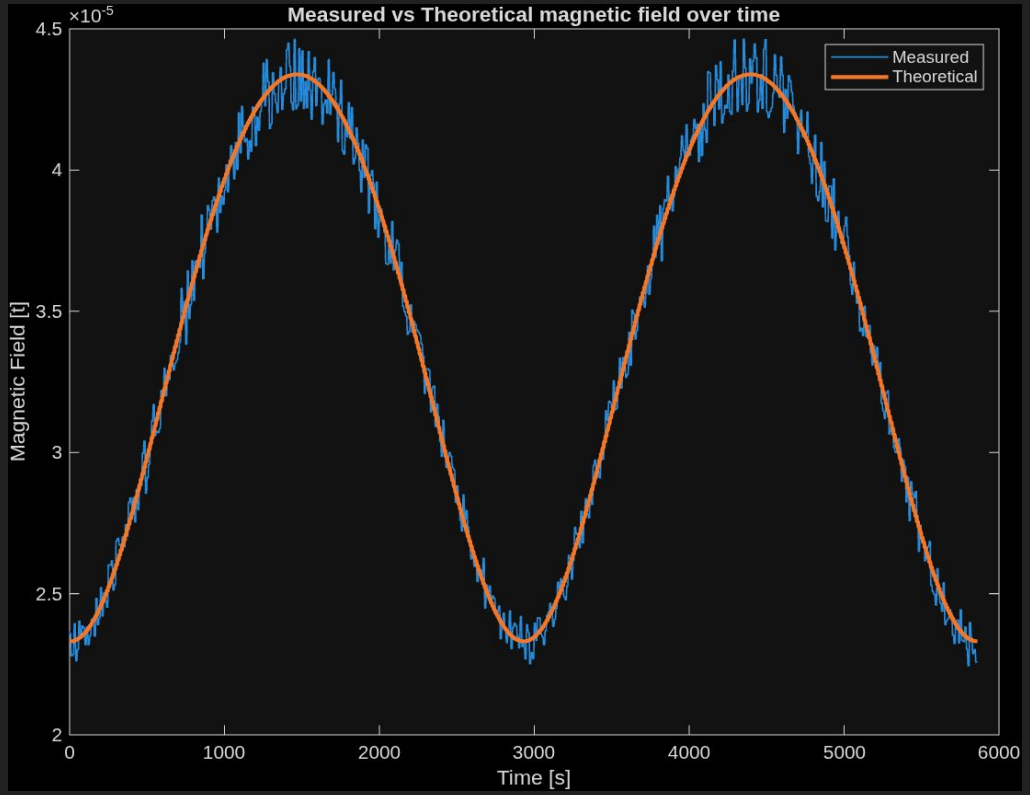
```
angular_velocity = [1e-4, 0.1, 1e-4];
inertia_tensor   = diag([0.2 0.4 0.6]);
```





Sensor noise

```
% Noise is 5% of the actual field,
% pointing in a random direction
noise_mag = norm(b_actual) * 0.05;
```



This is a crude noise function that is not used in further results

Detumbling

- Using a magnetic coil along each of 3 axises
- B-dot velocity estimation, enhanced B-dot law
- “The absolute angular speed can be reduced down to a value of the same order of magnitude as the orbit rate [i.e. mean motion]”

Avanzini, G., & Giulietti, F. (2012). Magnetic detumbling of a rigid spacecraft. Journal of Guidance, Control, and Dynamics, 35(4), 1326–1334. https://doi.org/10.2514/1.53074

$$k_\omega = 2\Omega(1 + \sin\xi_m)J_{\min}$$

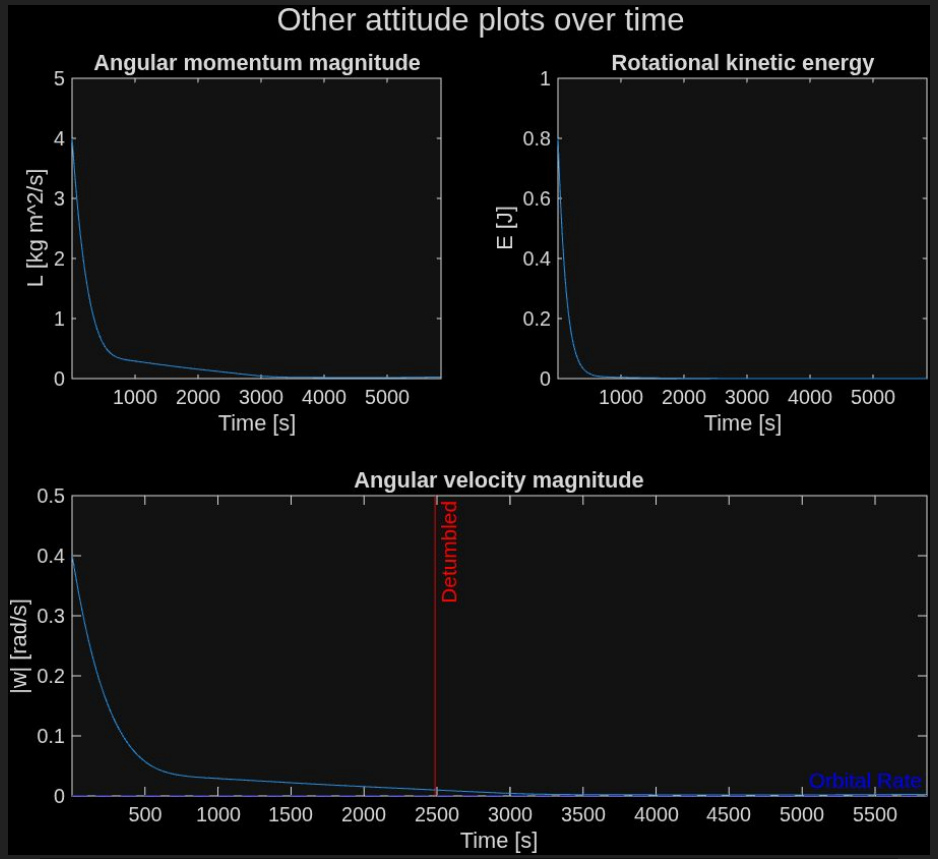
```
% Calculate control value K for magentotorquer
j_min = min(vecnorm(sat.Inertia_Tensor, 2, 2));
sat.K = (2*sat.Mean_Motion) * (1 + sin(sat.Inclination)) * j_min;
```

$$\boldsymbol{m} = -\frac{k_\omega}{\|\boldsymbol{b}\|}\dot{\hat{\boldsymbol{b}}}$$

```
sat.M_cmd = -(sat.K/norm(B_body)) * B_hat_dot;
```

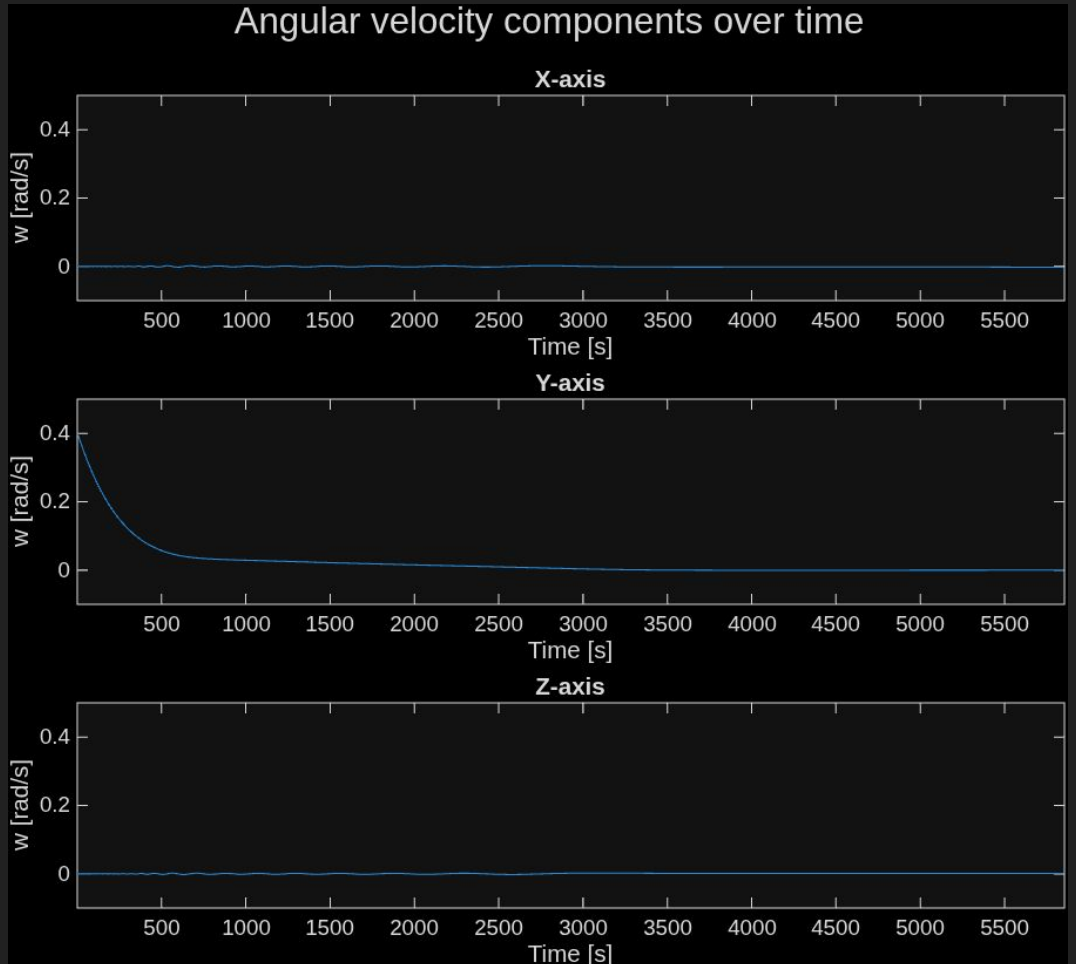
```
magnetorquer_torque = cross(sat.M_cmd, magnetic_field_body);
```

1-Axis Detumble: 1 orbit summary

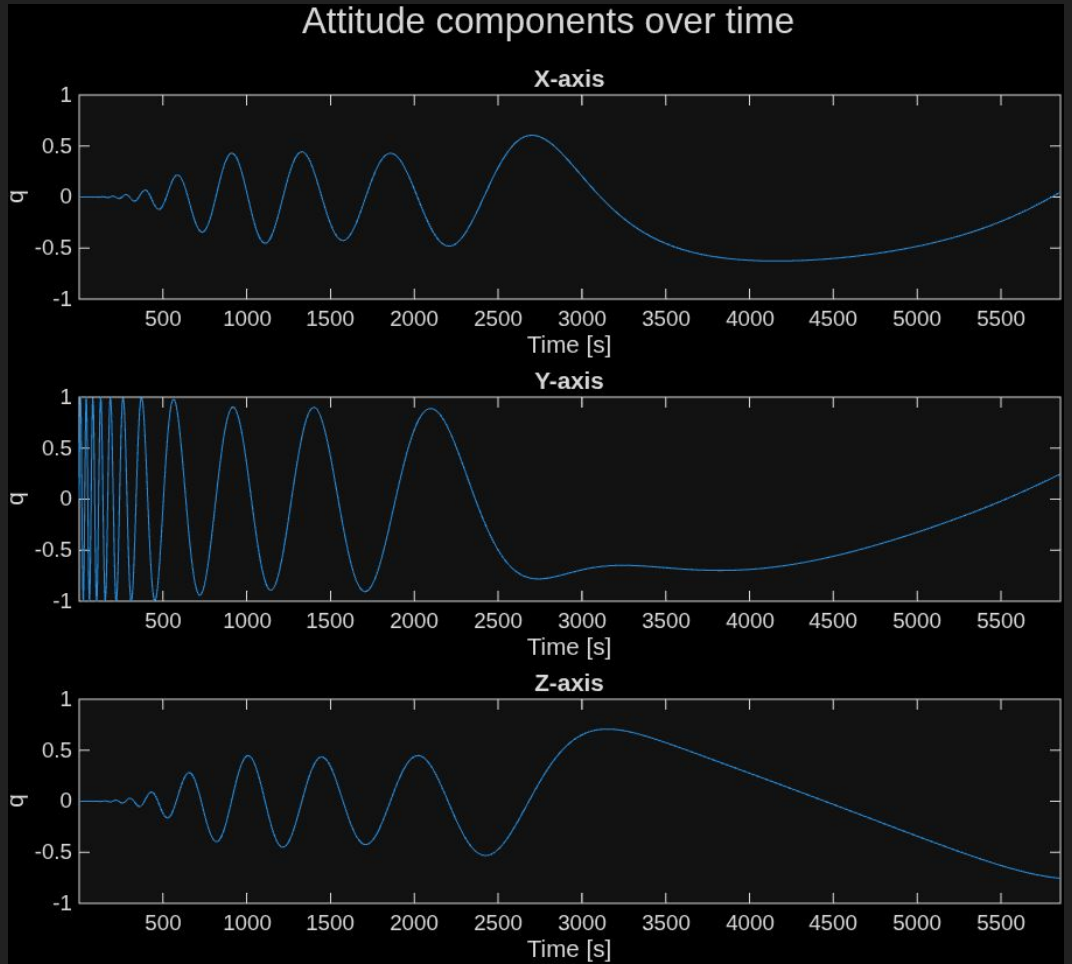



```
angular_velocity = [0, 0.4, 0];
inertia_tensor   = diag([10 10 10]);
```

1-Axis Detumble: Angular velocities



1-Axis Detumble: Attitude



3-Axis Detumble: Replicating paper results

Table 1 Spacecraft and orbit data, with initial conditions for cases A, B, and C

Parameter	Symbol	Value	Units
	Spacecraft and orbit data		
Principal moments of inertia	J_1, J_2, J_3	0.33, 0.37, 0.35	$kg \cdot m^2$
Maximum magnetic dipole	m_{max}	2.0	$A \cdot m^2$
Orbit radius	r_c	7021	km
Orbit period	T_{orb}	5855	s
Geomagnetic tilt angle	γ_m	11.44	deg
Inclination: case A	i	11	deg
Cases B and C		65	
	Initial conditions		
Cases A and B	$\boldsymbol{\omega}_0$	$(0.604, -0.760, -0.384)^T$	rad/s
	$\mathbf{Q}_0$	$(-0.062, 0.925, -0.007, 0.375)^T$	
Case C	$\boldsymbol{\omega}_0$	$(1.0, 0.0, 0.0)^T$	rad/s
	$\mathbf{Q}_0$	$(0.525, -0.514, 0.206, 0.646)^T$	

```
semi_major = 7021e3; % Meters
inclination = deg2rad(65); % Radians

angular_velocity = [0.604 -0.760 -0.384];
inertia_tensor   = diag([0.33 0.37 0.35]);
attitude = [-0.062 0.925 -0.007 0.375];
```

3-Axis Detumble: 3 orbits summary

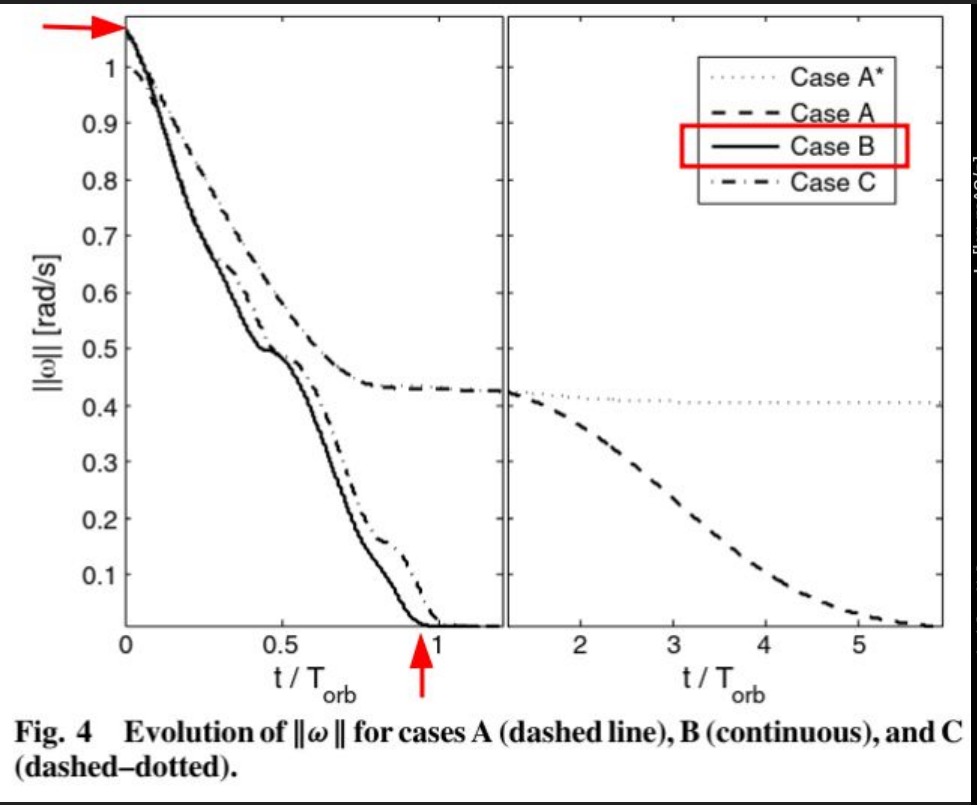
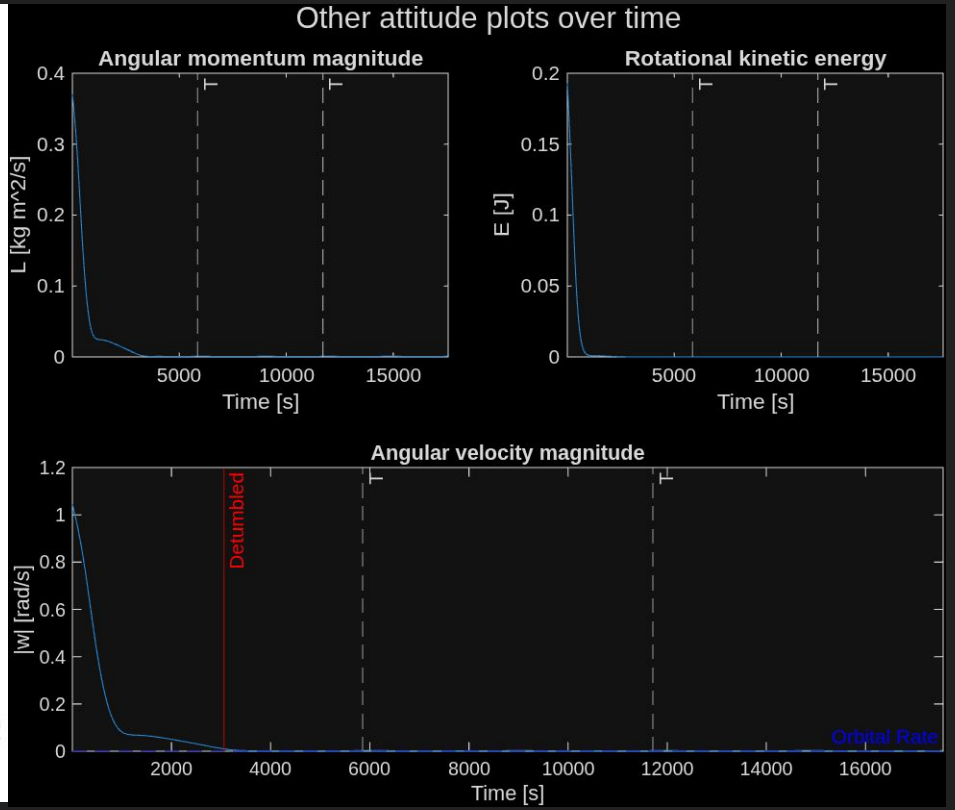
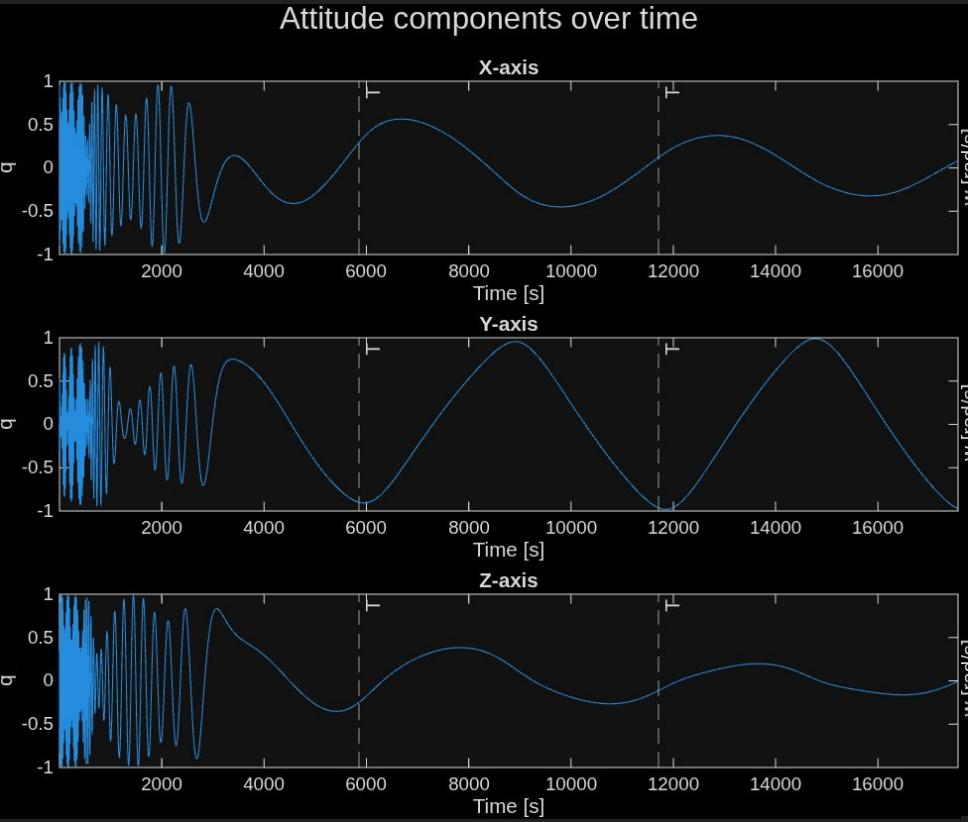


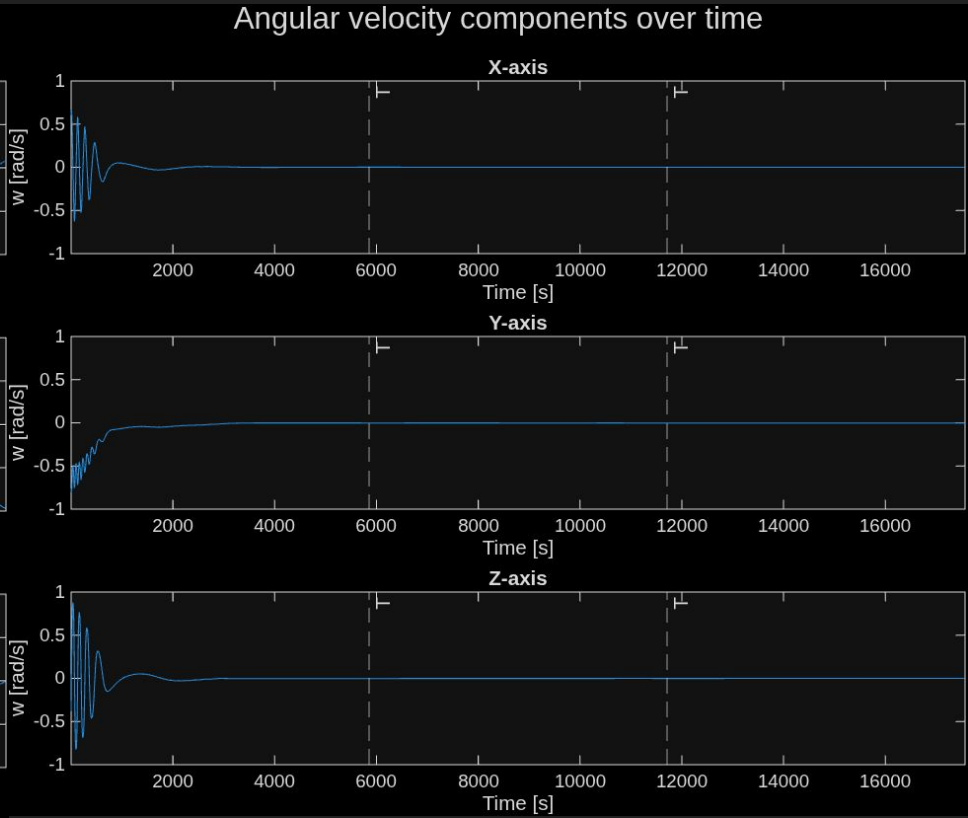
Fig. 4 Evolution of $\|\omega\|$ for cases A (dashed line), B (continuous), and C (dashed–dotted).




```
|w| down from 1.04398 rad/s to 0.00265 rad/s, which is of the same
order of magnitude as the satellite's Mean Motion 0.00107 rad/s
```

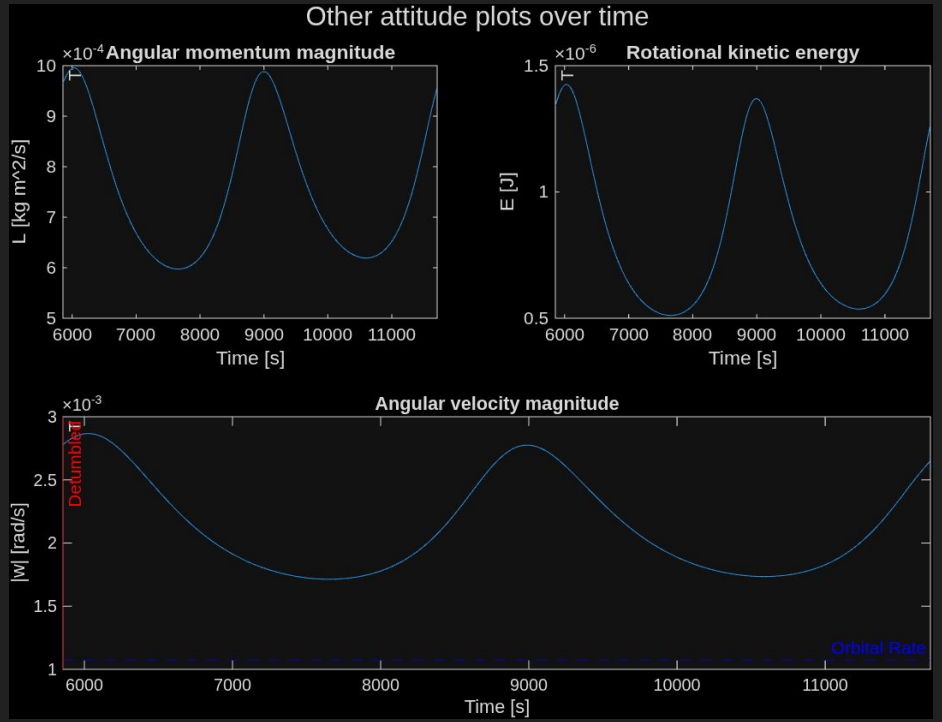
3-Axis Detumble: Attitude & Velocity





Zooming in

Since orbital motion can be interpreted as tumble by the controller, the B-dot command law reaches only a periodic steady-state - limit cycle - “featuring two oscillations per orbit”.

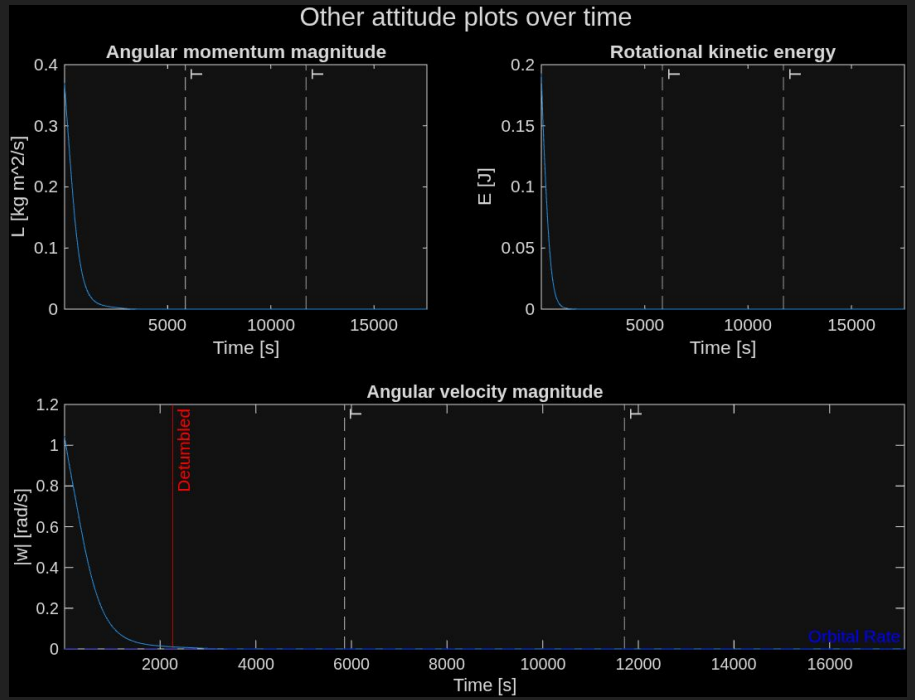


“The residual angular speed is related to the error in estimating the absolute time derivative of the magnetic field from onboard measurement only when no absolute attitude information is available”

3-Axis Detumble knowing angular velocity

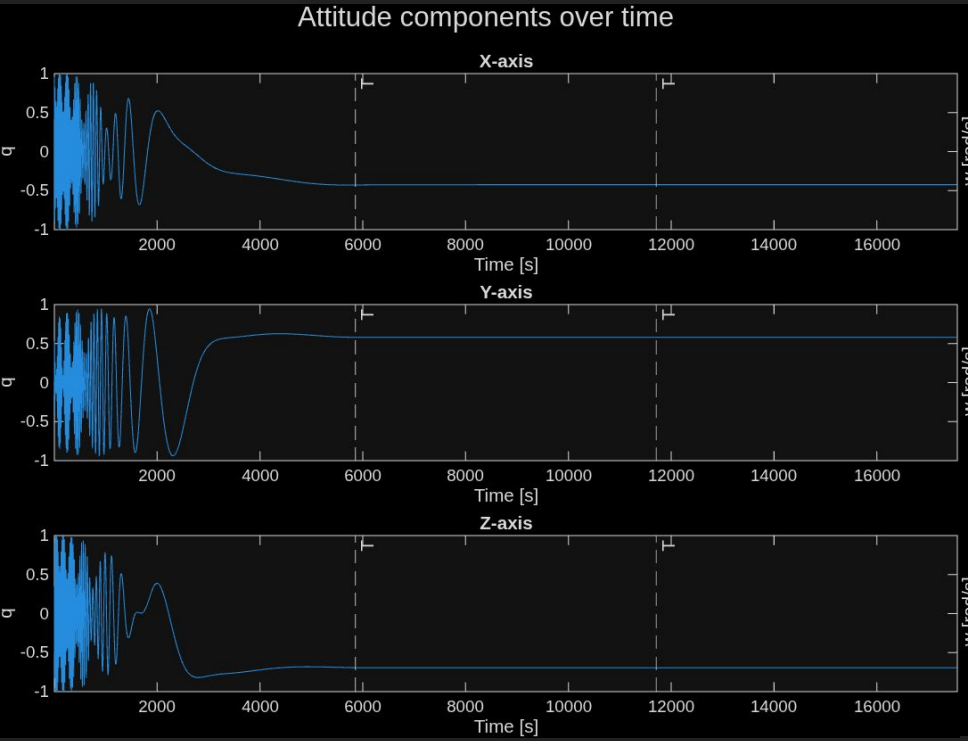
$$\boldsymbol{m} = -\frac{k_\omega}{\|\boldsymbol{b}\|}\hat{\boldsymbol{b}} \times [(\mathbb{I}_3 - \hat{\boldsymbol{b}}\hat{\boldsymbol{b}}^T)\boldsymbol{\omega}] = -\frac{k_\omega}{\|\boldsymbol{b}\|}(\hat{\boldsymbol{b}} \times \boldsymbol{\omega})$$

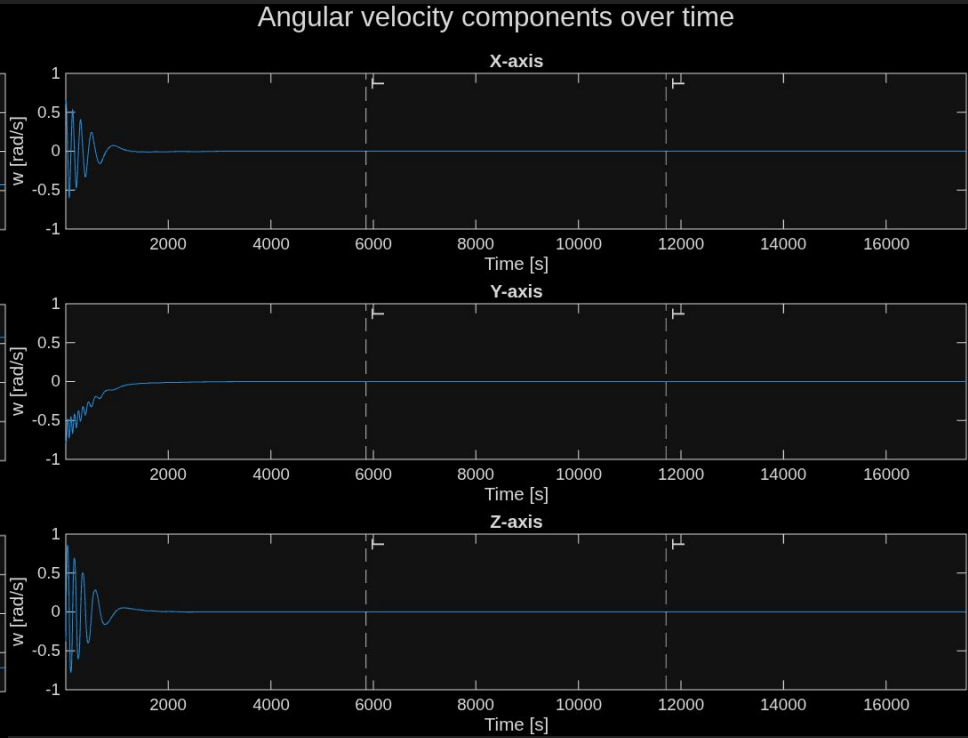
```
sat.M_cmd = -(sat.K/norm(B_body)) * cross(B_hat, angular_velocity);
```



|w| down from 1.04385 rad/s to 3.802931e-13 rad/s,
which is even less than the satellite's Mean Motion 0.00107 rad/s

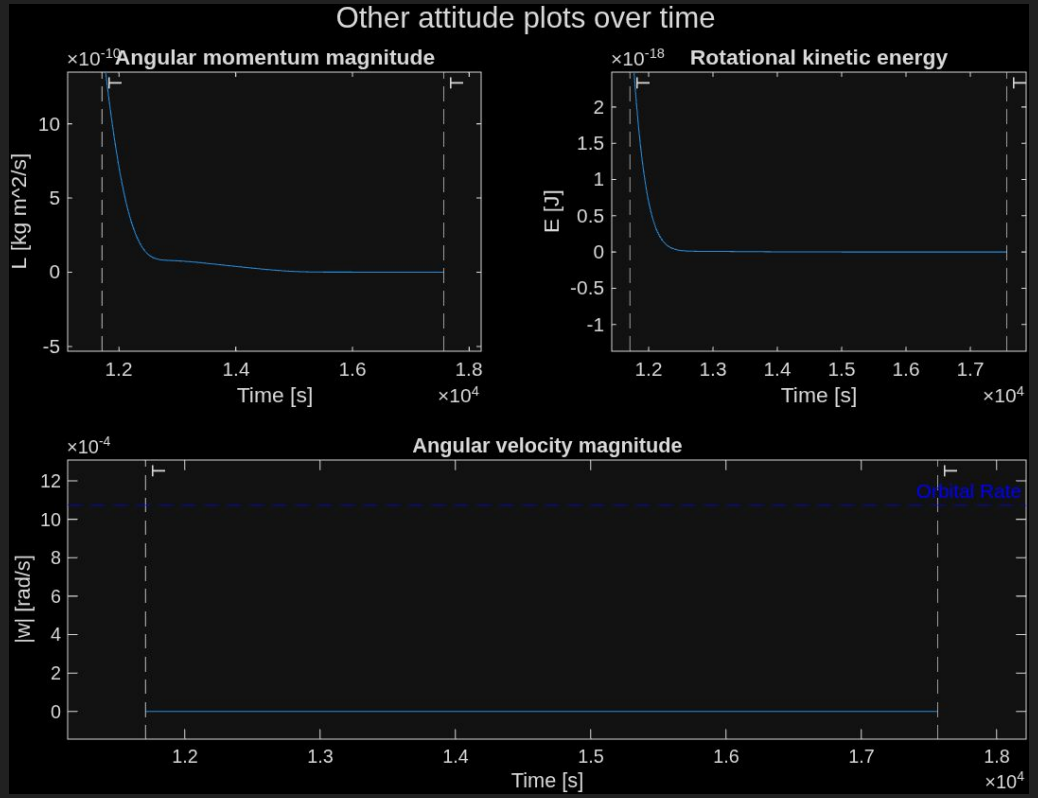
3-Axis Detumble knowing angular velocity





Zooming in knowing angular velocity

With perfect angular velocity information, the controller reaches a steady state.



Detumbling with less inclination

Table 1 Spacecraft and orbit data, with initial conditions for cases A, B, and C

Parameter	Symbol	Value	Units
	Spacecraft and orbit data		
Principal moments of inertia	J_1, J_2, J_3	0.33, 0.37, 0.35	$kg \cdot m^2$
Maximum magnetic dipole	m_{max}	2.0	$A \cdot m^2$
Orbit radius	r_c	7021	km
Orbit period	T_{orb}	5855	s
Geomagnetic tilt angle	γ_m	11.44	deg
Inclination: case A	i	11	deg
Cases B and C		65	
	Initial conditions		
Cases A and B	$\boldsymbol{\omega}_0$	$(0.604, -0.760, -0.384)^T$	rad/s
	$\mathbf{Q}_0$	$(-0.062, 0.925, -0.007, 0.375)^T$	
Case C	$\boldsymbol{\omega}_0$	$(1.0, 0.0, 0.0)^T$	rad/s
	$\mathbf{Q}_0$	$(0.525, -0.514, 0.206, 0.646)^T$	

```
semi_major = 7021e3; % Meters
inclination = deg2rad(11); % Radians

angular_velocity = [0.604 -0.760 -0.384];
inertia_tensor   = diag([0.33 0.37 0.35]);
attitude = [-0.062 0.925 -0.007 0.375];
```

Detumbling with less inclination

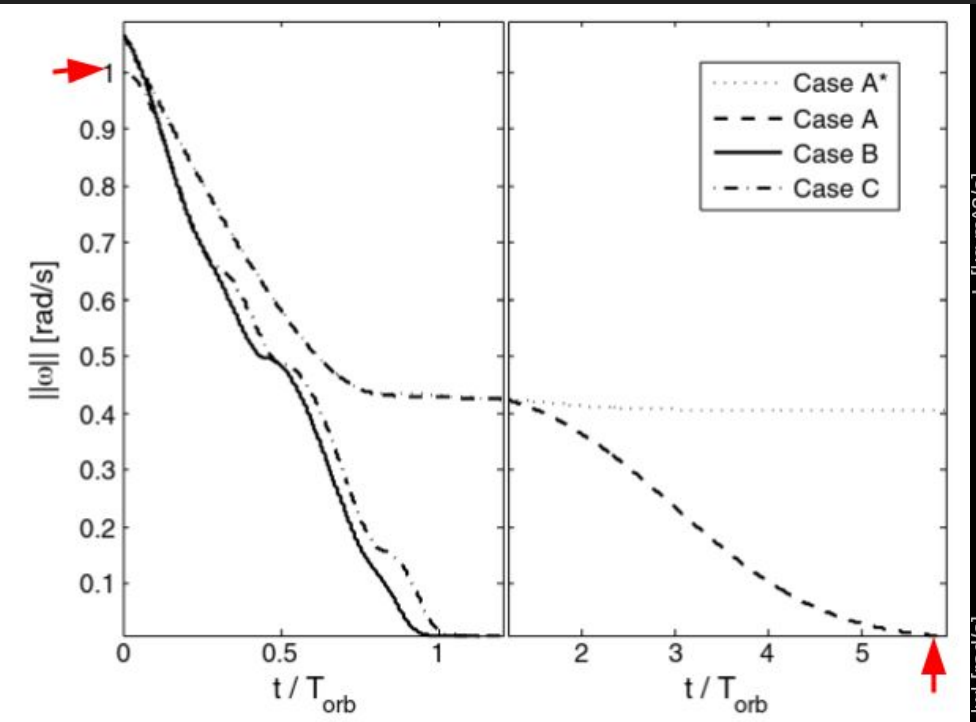
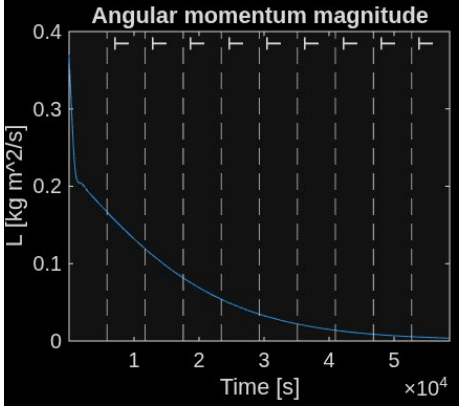
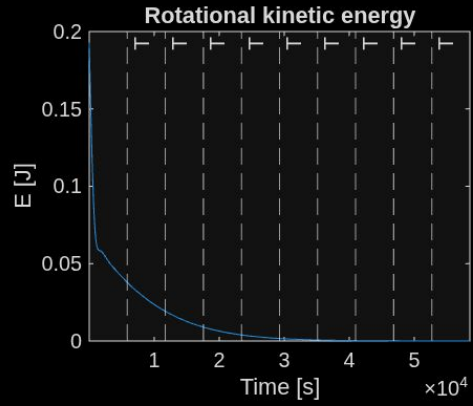
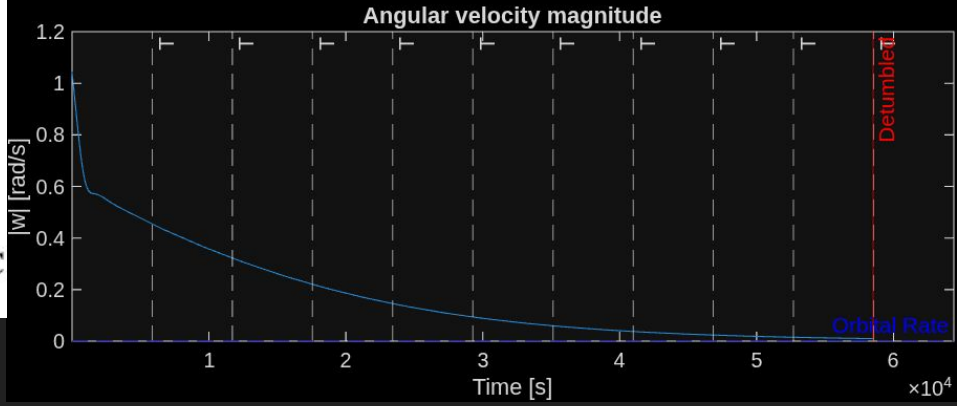


Fig. 4 Evolution of $\|\omega\|$ for cases A (dashed line), B (continuous), and C (dashed–dotted).

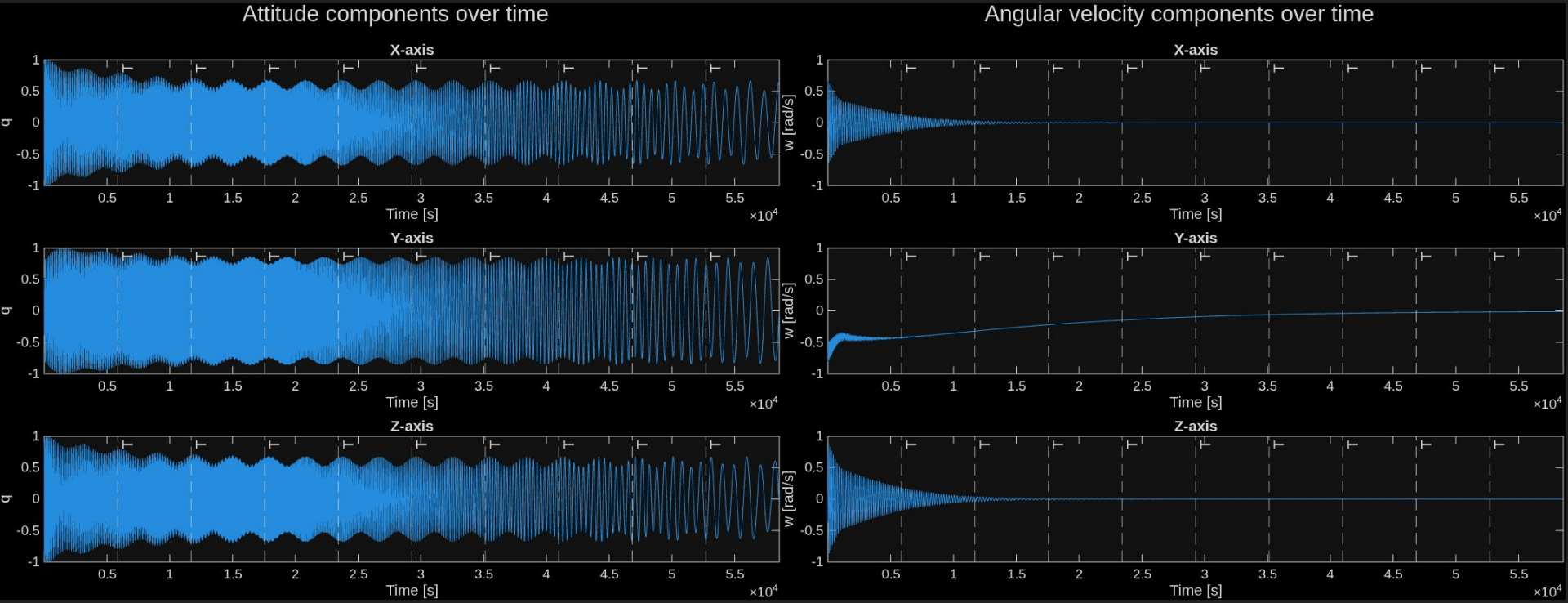
Other attitude plots over time







Detumbling with less inclination



Detumbling with less inclination

"...the rotation of the geomagnetic equatorial plane slowly varies the inclination of the orbit plane with respect to the geomagnetic equator"
"...more than five orbits are necessary [to detumble]"

In my model, the geomagnetic equatorial plane is fixed relative to the orbit plane, meaning the magnetic field direction varies less around the satellite, so detumbling takes an extra 5 orbits in the previous example.

Suggested Improvements

- Add planetary rotation that will vary geomagnetic equatorial plane
- Upgrade from dipole to magnetosphere
- Implement dynamic sampling frequency to improve simulation performance